

The Process of Statistical Modelling

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Introduction

Statistical modelling is the art of developing a statistical model to answer the **questions** the scientist want. It has to do with the *formulation* of a probabilistic model, the *estimation* of its parameters, its *validation*, its *comparison* with other relevant models, and its *interpretation*. Within the International Statistical Modelling Workshop, IWSM, a conference running for over 40 years surprisingly has been no attempts to define what we mean by *Statistical Modelling*. This situation remind us the statistical concept of *skewness*. Everyone know what we mean by skewness, but mathematically it is a very difficult concept to define. Statistical Modellers practice statistical modelling, they assume a model, fit it, compare it, check it and use it, but very few attempts are made to define the process and its components properly. In this paper, we attempt to do so. We do not expect to get everything right but we would like to start a debate about it. Hopefully this debate would lead to a better understanding of the statistical modelling process how to improve it and its pitfalls. We will argue that statistical modelling is a *process* composed of distinct *components*. The characterization of these components, would helps us to understanding better the structures of the process and to identify potential difficulties. In addition we would like to emphasize the concept of *focused statistical modelling* (FSM), that is, aligning model choice to the specific **purpose** of the study something, we feel that is overlooked in the literature. For example, it is not good to base your model comparison using mean square error which let you how good you fit the middle of the distribution of the response when the interest of study happens to be in the tail. As motivation and to help the discussion we will use a simple example of distributional regression, Stasinopoulos et al. (2024), in which reference curves are fitted and used to identify individuals at risk using the concept of z-scores.

Section ?? introduces the example. Section ?? introduces the basic concepts of SM. Conclusions drawn in section .

The data, the questions and the models

Over forty years ago **data analysis** was the realm of statisticians. While today statistician still play an important role, data analysis is the major subject of data scientists, numerical

analysts, data modellers, data miners, machine learners, and artificial intelligent researchers. They all used data to extract information about the world we are living in. What went wrong for the statistician? The answer is, partly, because mathematical statisticians who dominate the statistical community in the 80's were sort-sighted, see below, but also because computer power was expanding really fast the last few decades allowing computations that were unimaginable few years ago.

Most of the statistical departments forty years ago were occupied by mathematical statisticians. The leading statistician at the time were too arrogant to hear the questions that the researchers were asking and too busy to fight between themselves in the discussion between frequentist and Bayesian school of thoughts meanwhile paying little attention to the questions the scientists were asking. They were obsessed developing models and theory, with sound mathematical background but pretty unrealistic for practical purposes. They have fallen close to what now we called **Maslow's hammer**; "I suppose it is tempting, if the only tool you have is a hammer, to treat everything as if it were a nail." But not all statisticians were like that. The following remarkable quote appears in page 13 of Tukey J (1962) : "Far better an approximate answer to the right question, which is often vague, than an exact answer to the wrong question, which can always be made precise."

Mathematical statisticians were very keen to provide theorem, (most of them concerning asymptotic theory) which hardly answered practical questions. Few partitioners are caring on what happen if N , the number of observations, goes to infinity. People want to know what their data really tell them. In addition the problem with asymptotic theory itself is that no one knows at which points N is big enough to be considered that it reach its asymptotic behaviour? What is the length of a string? Bayesian statisticians, on the other hand, did try to address the real problems, see Chapter 1 and 2 of McElreath (2018), but at the time they were inhibited by their methodology. Monte Carlo Markov Chain (MCMC) a simulation computer intensive methodology, invented in the 40's rediscovered in late 80's was able to fit Bayesian models and become popular by the millennium by which time statistical modelling has move on to more complex algorithmic models. If you are using a "correct" model Bayesian inference is probably the best way to answer the questions about uncertainty. But how do you know you have the right model? We believe that "the type of statistical inference used (frequentist or Bayesian) is less important to the conclusion than choosing a suitable model in the first place". Note that in the above quote by conclusion we refer to answering the scientific **question** pose. In simple words, we feel, that using the right model is more important than using the right type of inference.

While the statistical models forty years ago were too simplistic to realistically, answer the **question** asked, the computer power was increased exponentially and according to Moore's law. Moore's law state that the number of transistors on an integrated circuit would roughly double every two years. Moore's law happen to be a very accurate description of reality until recently where while transistor density continues to improve the pace is slower. Moore's law allow the fitting of more complicated **models** and the fitting of larger **data** sets. The science of *data mining* was created as a way of extracting information from very large data sets. The same

time statistical models become less mathematical orientated by the introduction of what is now known as *algorithmic models*. The term *algorithmic models*, is linked to Leo Breiman, (Breiman 2003), although he did not use the exact phrase but instead he talked about “the algorithmic modelling culture...”. An algorithmic model is using algorithms to model unknown functions and treats the data-generating mechanism as unknown and focus on prediction (e.g., random forests, neural networks, boosting). Breiman contracted algorithmic models to “data models”: models based on probabilistic assumptions about the data-generating mechanism (e.g., linear regression, generalized linear models). We will refer to those models as *mathematical models* since they are based on mathematical assumptions and concepts.

As an example in our reasoning, about the statistical modelling process, we will use data from Judah *et al.* (2025), who provide reference ranges for liver function tests in a population of healthy pregnancies after adjusting for the effect of gestational age and other maternal demographic characteristics. Here, for simplicity, we will adjust only for gestational age, the most important factor. Therefore, we only have two (continuous) variables in the **data**: gestational age (**ga**) as the *covariate* and Albumin (**ALB**) as the *response*. The latter is an indicator of how well the liver functions. The data were obtained at different times during fetal scans, so they are cross-sectional data, that is, they are coming from 3429 different women. The data can be found in the the data set **ALB** in the R package **gamlss.data** and are shown in Figure 1

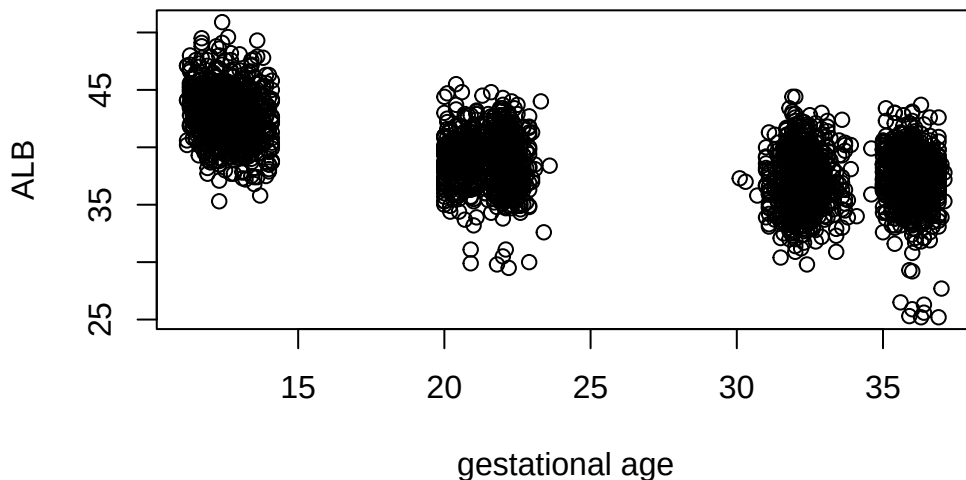


Figure 1: The ALB data: ALB against gestsion age. The gaps in the data are due to the fact that the collection time was during different scans times

The **purpose** of the study is to identify people with abnormally low ALB values given their gestational age, because they are at higher risk. This is a typical use of reference curves otherwise known as centile curve estimation. For example, WHO (2006) created standard reference curves whose purpose was to identify abnormalities in children’s growth. For our specific data set, the **question** is how to identify individuals with abnormally low ALB values. In other words, we are interested at the extreme left tail of the conditional distribution of ALB given **ga** and this is purpose of the study. If we know the *population* quantile/centile values

for **ALB** we could identify individuals at the extreme left tail of the distribution of **ALB** given **ga**, e.g. a value lower than the 2.5% centile value of the **ALB** distribution given **ga**. Since we only have a *sample* from the *population* (the **data**) and in order to make *inferences* about the population, (to go from the specific to the general) we need a **statistical model**. More specifically a distributional regression model, see Section , could answer our **question**.

The discussion above had identify three components of the statistical modelling process.

- The **question**: what the scientist would like to answer
- The **data**: the source of information to answer the question and
- the **model**: which is the filter to extract the right information from the data

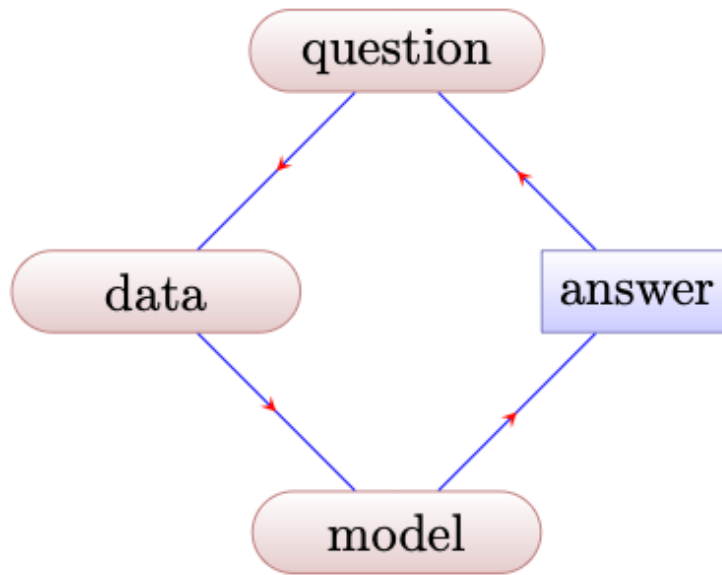


Figure 2: The question data and model interrelationship

A important point in our discussion is the interconnection of the above three components. shown in Figure 2. The data are collected to answer the question. Occasionally, the data already exist prompt the question(s). The model concentrates to the most important information in the data in order to answer the question. At this stage we have to ask ourself the following questions before embark on any statistical modelling;

- is my question clear enough to solve the scientific argument?
- is my data reliable and capable to answer the question?
- is the model the right one to answer the question?

Data reliability is something that pops out very often in any scientific discussion but it is too big a subject to discuss here. There is an aspect in the data which of important in any

statistical modelling process with mentioning here. It has to do with the structure of the data and whether the data are independent from each other or not.

For **tabular data**, that is, data which can be represented as data matrix, [columns as variable and rows as observations], spreadsheet type, independence means that the rows of the data are not related in any way between them. Exceptions to that are;

- **time series** data where subsequence values depend on the previous values
- **spatial** data where values of variable depend on the values of their neighbours
- **cluster** data where there is a within cluster correlation while clusters are independent from each other.

All the above features of the data must be part of the statistical modelling process. Let us concentrate now on models.

The model

We are firmly believers of the George Box (1979) quote;

all models are wrong but some are useful.

We believe that there is no true model. The models are there to help us understand the reality. They are not reality. In fact the scientific consensus is that our brain uses models. Those models were developed during our evolutionary process to cope with the outside world (please give reference here). That is, what we think reality is it is what our brain model tell us. Therefore how useful a model is depends on its purpose. The models in our brains are good for our evolutionary survival, statistical model are useful whether they could answer the **question** of the study.

the regression model

We would like to concentrate on **regression** type of models only since they represent more than 90% of statistical modelling. Regression models are **input-output** models, Figure 3, where the input represent explanatory variable(s) X and the output the response (target) variable(s), Y . We are trying to model the response Y given X . In mathematical terms we are trying to model the response variable Y as a function of X . Regression model are *supervised* model since a target variable, always exist.

Let follow Breiman (2003) arguments in trying to describe how a regression model is functioning. We believe that the X explanatory variables (covariates) affect the response Y in a way too complex to understand,

$$X \longrightarrow \boxed{\text{NATURE}} \longrightarrow Y,$$



Figure 3: Input-output model

that because nature itself is too complex. In order to understand better what is going on we need a **model**.

$$X \longrightarrow \boxed{\text{Model}} \longrightarrow Y,$$

So a model is a mechanism on how X affects Y . Mathematically we could be represented a model by an unknown function $G()$;

$$X \longrightarrow \boxed{G()} \longrightarrow Y.$$

The task of a statistical modeller is to try to find this unknown function $G()$. This can be achieved using **mathematical**, or **algorithmic** models. A mathematical model, assumes that the function $G(X)$ is behaving in a certain mathematical way, for example using a linearly equation, or a partial differential equations, (PDE). The behaviour of $G()$ is defined through the parameters involve (which here we refer to as the **coefficients** or β 's). If we knew the β 's we know the structure of the model. So *estimation* of $G()$ in a mathematical model involves finding the best set of β 's given the data. An algorithmic models defines the function $G()$ algorithmically. The algorithmic specification of $G()$ still involves a mathematical structures but typically the $G()$ is over-parametrized (it contains too many β 's) and we need a set of **hyper-parameters**, λ to control its behaviour. The function $G(X)$ then is estimated from the data by identify the best β 's and λ 's.

In statistical modelling the model is not completely *deterministic*, that is, $Y = G(X)$ is not sufficient to model Y because there exist a **natural error** variation associated with Y . The reasoning behind this natural error variation come from the observation that if we could repeat the experiment several times even with identical X values, the output values of the response Y will be somehow slightly different. This difference in behaviour of the response is the natural error associated with Y . In statistical terms Y has a distribution. In fact, a conditional distribution $\mathcal{D}(Y|X)$ depending of the values of X . It is common to call the function $G()$ the **signal** while the natural error distribution $\mathcal{D}(Y|X)$ the **noise**. In order to complete the specification of a regression model the signal, $G()$, on its own it is not sufficient we also need the specification of the noise, the stochastic (probabilistic) component of the model. Therefore a minimum notation of any regression type model $\mathcal{M}$ requires at least two components; a *mathematical* component, G , describing the relation of between the input and output and a *stochastic* component, $\mathcal{D}$ describing the natural variation of Y , for example, $\mathcal{M}(G, \mathcal{D})$.

Assumptions

How those two components G and $\mathcal{D}$ are defined on a specific study depends of the set of **assumptions** the researcher is prepared to make. An **assumption** is an axiomatic statement which need to be approximately correct for a model to work otherwise the model is not useful. All models are based on assumptions but unfortunately not all assumptions are stated **explicitly**. Considered for example the common linear regression model used in statistics;

$$Y = \mathbf{X}\beta + \mathcal{E}. \quad (1)$$

Here the generic function $G()$ is replaced by a linear equation while the natural error distribution $\mathcal{D}(Y|X)$ by the **error**, $\mathcal{E}$, which is a vector of values assumed to be independent from each other and normally distributed with zero mean and constant variance. The linear function assumption of the model and the normality assumption of the errors in Equation 1 are **explicit** assumptions. Explicit assumptions are easily checked using **diagnostics** and the checking is often part of the statistical modelling process. For example the assumption of normality can be easily checked using a QQ-plot but often the model contains **implicit** assumptions. For example, in Equation 1, one implicit assumptions is that the signal $\mathbf{X}\beta$ and the noise $\mathcal{E}$ are **additive**. Another implicit assumption, it become obvious if we try to replace the a normal distribution with an asymmetric distribution. The error, $\mathcal{E}$, in Equation 1 needs a symmetric distributions around its mean. This is a consequence of the notation we used to define the model. The problem could be that the additivity of the signal and noise not appear in the checking list of a statistical modelling process so it is never checked. This is typical to all implicit assumptions they never checked properly. Another point concerning assumptions, worth mentioned here, is that assumptions often help with the **interpretation** of the model, that is, helping us answer the **question**.

Distributional Regression

If the linear function $\mathbf{X}\beta$ in Equation 1 is replaced with the generic function $G()$ we have the model

$$Y = G(\mathbf{X}) + \mathcal{E}. \quad (2)$$

a common form of a regression model used in *machine learning* where usually the construction of $G()$ is done algorithmic. For example, $G()$ is calculated using random forest, boosting or deep learning models. In Breiman's notation Equation 1 can be written as;

$$X \rightarrow \boxed{G(X)} \rightarrow \mathbb{E}[Y|X], \quad (3)$$

where $E[Y|X]$ denote the conditional expectation (mean) of Y given X . We feel then even the more general model in Equation 2 or Equation 3 is too restrictive. It deals on how the mean (or the location) parameter of the distribution of the response is behaving but say

nothing for other parts of the distribution for the response, like the variance, skewness or tails. If we are trying to model the general behaviour of the response we need a **distributional regression** model. In a distributional regression model we assume that the specification of the conditional distribution of the response depends on a set of parameters $\theta = (\theta_1, \theta_2, \dots, \theta_k)$, which correspond to properties of the distribution (i.e. mean, variance, skewness) and that **all** of the parameters could be a function of the explanatory variables i.e. $\theta = G(X)$. In Breiman's notation a distributional regression can be represented as

$$X \longrightarrow \boxed{G(X)} \longrightarrow \mathcal{D}(Y|\theta = G(X)),$$

where $\mathcal{D}(Y|\theta)$ represent the natural error distribution of Y . The first serious attempt for distributional regression is the Generalized Additive Models for Location, Scale and Shape, GAMLSS, introduced by Rigby and Stasinopoulos (2005), where the function $G()$ was assumed to be a set of additive possible smooth functions of the explanatory variables and the conditional distribution $\mathcal{D}(Y|X)$ to be **any** theoretical probability distribution function (p.d.f.). The set of p.d.f.'s including continuous, discrete or mixed variable distributions. In a GAMLSS model the task of the researcher becomes to find;

- **which** and **how** the X 's effect the parameters of the distribution, that is, to identify the structure of $\theta = G(X)$ and
- the **appropriate** conditional distribution for the response i.e. $\mathcal{D}(Y|\theta)$.

Note that the contribution of the explanatory variables come through the parameters of the distribution for the response therefore for interpretation ease the chosen parametrization of the distribution should modelled a specific characteristic of the distribution like for example the skewness.

💡 Mikis is up to here

A **modern** computational approach of the distributional regression, (**chai2026neural?**);

$$X, U \longrightarrow \boxed{\text{deep learning}} \longrightarrow D(Y|X),$$

where U represent a simple distribution e.g. uniform which after transformations becomes $D(Y|X)$ the conditional distribution of Y given X .

This version of the DR model is NOT easy to interpret because it is a **black box**.

Note that at any stage here we make the distinction about *predictive* and *data* based models. The reason is that we feel it is not helpful. A model is a model how to fit depends on the

fit

To fit a model we need a measure of goodness of fit

measurement of goodness of fit

A measurement of goodness of fit allows us to fit a model and also to compare to other models

diagnostics

The first statistical interactive computer program was GLIM introduced in the middle 1970's to fit Generalised Linear Models ()

The Albumin (ALB) data

The Albumin (ALB) (model)

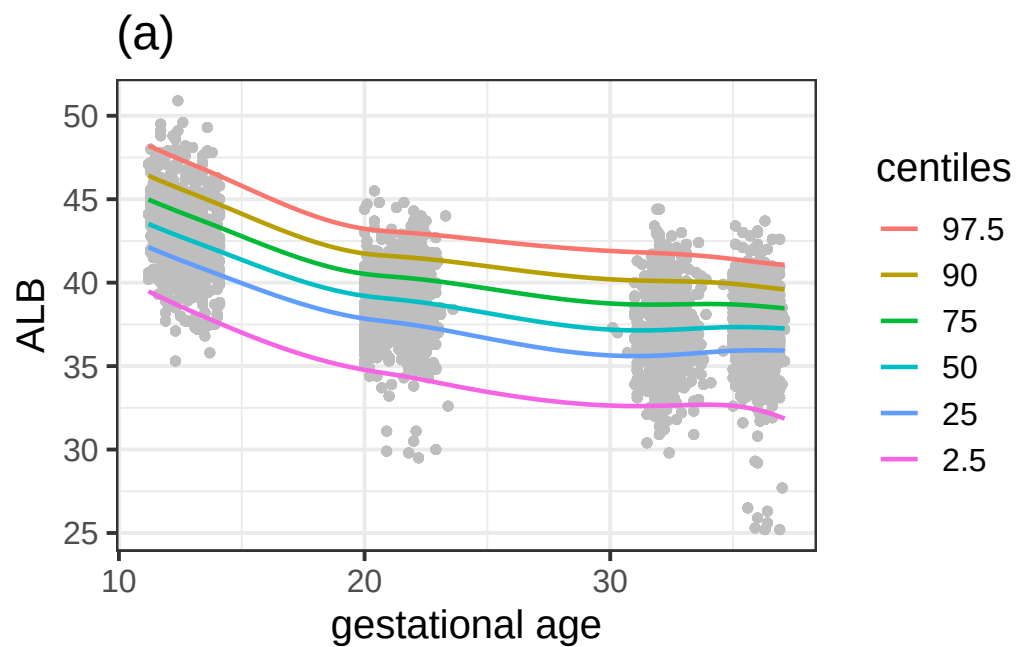


Figure 4

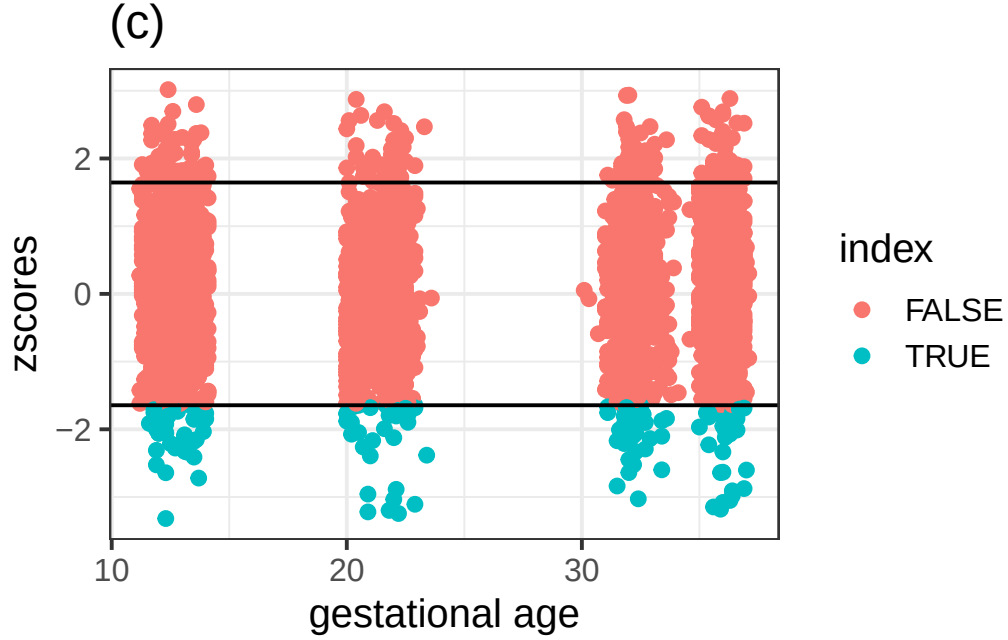


Figure 5

The Albumin (ALB) (question)

The **purpose** of the study is to identify people with abnormally low ALB values given their gestational age, because they are at higher risk. This is a typical use of reference curves. For example, WHO (2006) created reference curves whose purpose was to identify abnormalities in children's growth. For our specific data set, the **question** is how to identify individuals with abnormally low ALB values. In other words, we are interested at the extreme left tail of the conditional distribution of ALB given **ga**. If we know the *population* quantile/centile values for ALB we could identify individuals at the extreme left tail of the distribution of ALB given **ga**, e.g. a value lower than the 2.5% centile value of the ALB distribution given **ga**. Since we only have a *sample* from the *population* (our data) and in order to make *inferences* about the population, we need a **statistical model**.

A statistical model combines a *mathematical* or *algorithmic* component with a *stochastic* component. The first component helps to model the *signal* in the **data**, while the stochastic component helps to model the distribution around the signal. Note that in machine learning a typical algorithmic model e.g. a random forest, uses normal distribution errors with constant variance. Here because of the nature of the ALB we will use a *distributional regression* model, DR, because it has more flexibility to correctly model the tails of the distribution.

A DR is a statistical model where a distribution is **assumed** for the *response* variable given the covariates and where all the parameters of the distribution are potentially influenced by the covariates. There are two major statistical modelling parts in any DR: i) how to choose the

distribution for the response, and ii) how and which of the covariates affect the distribution parameters. In most occasions we start with the first, by *assuming* a parametric distribution for the response and fitting different distributions (using a reasonable model for the parameters), to find an adequate distribution for the response. Given a selected distribution, we then try to identify how the parameters are affected and by which covariates, e.g are the effects linear, non-linear, and how the covariates interact with each other. In our case, because we have only one co-variate, gestation age, we use a smoother for **ga** for each distribution parameter. A smoother is, a non-parametric smooth function which can capture non-nonlinear relationships (but which also at the limit could identify a linear relationship). Note that in our example, the second part of the DR analysis, on how to determine the affect of covariates to the parameters is rather trivial, but more generally when we have multiple covariates often it is the most difficult part of the analysis.

In conclusion, the main task of the current study is to find a suitable conditional distribution for ALB, given the gestational age, where the effect of gestational age is captured by a smoother. For example, for a four parameter distribution with parameters say, μ , σ , ν and τ we use model with smooth non-parametric functions $s(ga)$ for their predictors, $\mu : s(ga)$, $\sigma : s(ga)$, $\nu : s(ga)$, $\tau : s(ga)$, where the notation $s()$ represents a suitable smoother, for example, a P-spline (Eilers and Marx, 1996).

To choose between different distributions (having the same range of values as the response) we have used two different **measures of fit**. The first is log-likelihood based, while the second is a focus criterion based on χ^2 , see later. Both criteria are **test** measures, because both are evaluated at cross validated data (using 10-fold cross-validation explained later). The first criterion, the deviance, $D(\hat{\theta}) = -2\ell(\hat{\theta})$ (where $\ell()$ is the log-likelihood) is a global measure of fit is also used to **train** the model while the second is a focus measure more related to the question, of identification of individuals with low ALB. Both criteria selected the **BCTo** distribution. In fact, the test deviance select the **BCTo** distribution as best with **BCT** as second while the test focus criterion selected **BCT** as best with **BCTo** as second. The distributions are almost identical with **BCTo** using a log link for the parameter μ while **BCT** using an identity link.

The **BCTo** model had good test *residual diagnostic* plots. Figure (b) shows a *worm* plot, van Buuren and Fredriks (2001) of its training residuals. It is recommended that multiple worm plots of the residuals split into intervals by gestation age should be used but they are not shown here because of limitation of space. The **BCTo** distribution has four parameters all defined on the range $(0, \infty)$, where μ approximates the median of the distribution, σ approximates the coefficient of variation, and ν and τ are skewness and the kurtosis parameters, respectively.

The last but probably the most important part of statistical modelling is the **interpretation** of the model. Figure 1 (c) shows the z-scores (the training residuals) of the fitted DR model. Given the gestation age the z-scores can be calculated for any new value of ALB (even for the regions of gestational age where we do not have any data, although this can be dangerous as it assumes that the model is still valid in these regions). The final plot Figure 1 (d) is

to demonstrate the flexibility of a DR model and shows the shape of four different fitted distributions of ALB given four values of gestational age, one from each of the our four groups of gestational age.

Summarizing, we have the **data**, the **purpose** and we will try to build a **model** to answer the **questions**. We used cross-validated data and used appropriate **measures of goodness of fit** and **diagnostics** to select appropriate models and then try to **interpret** the model. Next we try formalize these concepts.

The basic components of statistical modelling

Statistical modelling is the *art* of creating a statistical model to represent the response variable generating process,

%It is a *filter* which extracts information from the data %e.g. **data** $\rightarrow$ **model** $\rightarrow$ **information**, either to make inferences about the *population*, %(a process known as *statistical inference*), or to gain knowledge from the information in the data. %, (a process known as *data analysis*). The aim is to answer the **questions** of the research which is also the **purpose** of the study. The first question to answer is *whether the data or the model are capable of answering the questions*.

Figure ?? in blue shows the classical statistical modelling process, introduced in 1970's in the first edition of time-series book by Box and Jenkins see Box *et al.* (2015). All data at this stage are treated as training data. The assumptions are repeatedly updated in order to obtain a model with adequate fit. Assumptions can be *explicit* or *implicit*. For example, mathematical models have mostly explicit assumptions while algorithmic stochastic models mostly implicit. Explicit assumptions are easier to check using model **diagnostics** tools. All models are built on assumptions even the so called “non-parametric” or “distribution free”. Often it is easier to make the assumptions and check them with diagnostics than trying to check whether the non-parametric or distribution free models are appropriate or not.

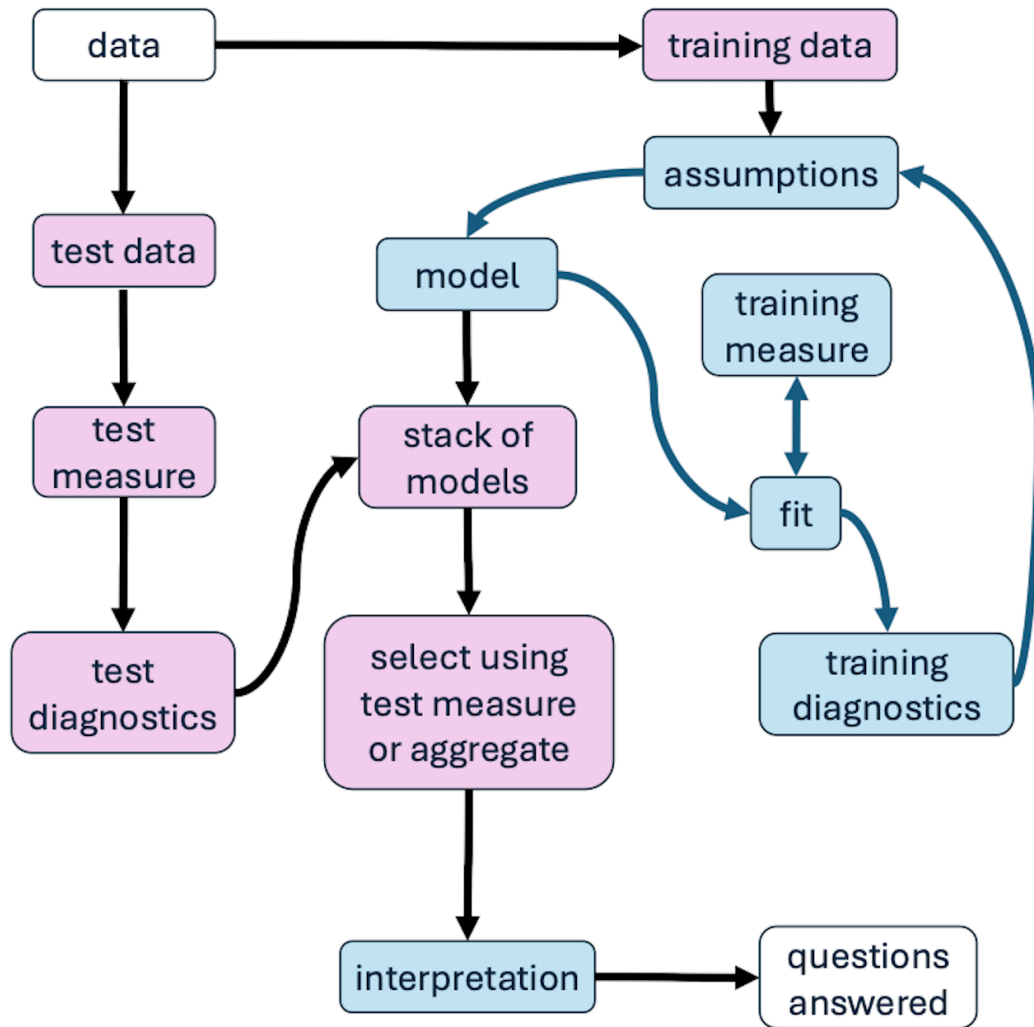
The training measure in classical SM is usually an appropriate objective **measure of goodness of fit**, (e.g., the fitted log likelihood).

The problems with classical SM are, firstly, that it aims to find an single overall adequately fitted model avoiding underfitting, but does not check for overfitting. Secondly, by looking at the overall fit, it does not necessarily focus on the **purpose** of the study. In the ALB example the purpose is to identify abnormally low values of ALB given **ga**. Hence the fit to the lower tail of the distribution of ALB given **ga** matters more than the overall fit.

A more modern SM approach is shown in Figure ?? in blue and purple. In order to find a good model for the response variable, we require an adequate model (i.e. avoid underfitting), but also avoid overfitting. In order to find possible models that ensure both, we need to create **test data** to validate the model. For relatively small data sets we recommend a 10-fold cross-validation, followed by diagnostic checking of the combined test residuals from all 10 folds. Finally to compare possible models, we apply focused statistical modelling (FSM), which

creates a **test measure** (appropriate to answer the **questions** of the study), a function of the observed response variable values and their corresponding 10-fold fitted response variable test distributions. For the ALB example in Section ??, our proposed FSM test measure is the value of a Chi-squared test statistic, number of observations (in a 2x4 table, with rows defined by above or below the 5% centile, and columns defined by splitting the data into the 4 gestational age intervals). %Model Selection? Overfitting?

Modern Statistical Modelling



Conclusions

We have compared the classical statistical modelling approach with more modern approaches using a simple distributional regression model. We recommend selecting models based on the combined test data from a 10-fold cross-validation. We recommend choosing models using a focused test data objective measure (when available) related to the purpose or objective of the study. For example, it is not appropriate to use an MSE criterion, which is suitable for evaluating the fit in the center of the distribution, to assess predictive performance when the interest lies in the behavior of the tails of the distribution. To describe the connection between the focused test measure and the underlying purpose or research questions of a study, we introduce here the term **focused statistical modelling** (FSM).

%*****

% Tables can be included at any place in the text. % As general format, use tables with horizontal lines only % (i.e., no vertical lines separating columns). % Start and end each table with a double horizontal line. % Tables are incorporated using a table environment:

% In the text, reference to the Table can be made as follows: %We refer to Table~?? for a summary of our main results. Have a look to Table~?? for %an additional example.

%*****

% Figures can be included at any place in the text. % The only allowed formats for figures are pdf files. % % Please, do not include figures in any other format. % % Use file names which are very unlikely to be used by % other contributors; for example, use file names starting % with the surname of the first author. % Figures are incorporated using a figure environment: % Make sure you specify the extension of the file (pdf)

%Finally a figure (in .pdf!)

% In the text, reference to the Figure can be made as follows: %We refer to Figure~?? for a~graphical representation.

%*****

Acknowledgments,

if needed:

Glossary

references

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